

Pharmaceutical Care Program for Patients With Uncontrolled Hypertension

Report of a Double-Blind Clinical Trial With Ambulatory Blood Pressure Monitoring

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Background: Pharmaceutical care programs may be an option to improve blood pressure (BP) control in patients with uncontrolled hypertension. The aim of this study was to evaluate the efficacy of pharmaceutical care programs in treating patients with resistant hypertension.

Methods: In a double-blind randomized clinical trial, 71 patients with uncontrolled BP were enrolled in a pharmaceutical care program or in a control group and underwent a series of cognitive tests. The primary outcome was change in ambulatory BP (ABP) between the baseline evaluation and the final visit 6 months later. The secondary outcomes were the frequency of drug-related problems and adherence as determined by plasma levels of hydrochlorothiazide.

Results: The δ -values between the intervention and control groups for ABP in the different daily periods, with

the corresponding 95% confidence limits, adjusted for age and baseline BP were: 3 (–1 to 5), 2 (–2 to 4), and 5 (–1 to 6) mm Hg for 24 h, daily and nightly systolic BP, respectively. The corresponding values for diastolic BP were 1 (–1 to 3), 0 (–2 to 2), and 3 (–1 to 4) mm Hg, respectively. Hydrochlorothiazide was detected in the plasma in 21 of 27 patients in the intervention group that attended to all appointments and 24 of 30 patients in the control group ($P = .904$).

Conclusions: The pharmaceutical care program tested in this trial was feasible and showed a trend for better BP control in patients with uncontrolled hypertension. Am J Hypertens 2006;19:528–533 © 2006 American Journal of Hypertension, Ltd.

Key Words: Hypertension, uncontrolled blood pressure, pharmaceutical care.

Hypertension is among the leading causes of cardiovascular disease worldwide.¹ Regardless of the availability of many efficacious drug and nondrug treatments, the rate of control among populations is far from ideal.² Inertia of physicians in the face of uncontrolled blood pressure (BP), adverse effects of BP-lowering drugs, attitudes of patients in relation to disease and treatments, and lack of structured, efficient health care facilities are some reasons for the low rates of hypertension control.^{3,4}

Pharmaceutical care is an innovative approach designed to improve the quality of care of both ambulatory and hospitalized patients.^{5,6} Its aim is the prevention, identification,

and control of drug-related problems. Randomized clinical trials that evaluated the performance of clinical pharmacist services showed improvement in the adherence to treatments, in the understanding of dosage schedules and allowed the early recognition of adverse effects.^{7,8} Improvement of the rates of control of chronic diseases has been reported with pharmaceutical care,^{9,10} but its effectiveness in the management of patients with hypertension has been scarcely tested, and the studies present methodologic shortcomings.^{7,11}

In this report we present the results of a randomized clinical trial that tested the efficacy of pharmaceutical

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intervention to achieve better BP control in patients with uncontrolled hypertension.

Methods

Setting, Patients, and Study Design

The study was carried out in the Hypertension Clinic of the Hospital de Clínicas de Porto Alegre (Porto Alegre, RS, Brazil).¹²⁻¹⁴ Adult patients (≥ 18 years of age) who were receiving treatment for hypertension at least with hydrochlorothiazide, with uncontrolled BP (average of two office BP measurements $\geq 140/90$ mm Hg), were invited to participate in a double-blind, randomized clinical trial with evaluation of BP by ambulatory BP (ABP) monitoring. The random allocation was done in blocks of eight patients each and stratified by gender through a computer-generated sequence. Patients were followed-up for 6 months. Figure 1 gives an overview of the patient groups at various stages of the trial.

The Institution Review Board approved the investigation, and all patients gave their written consent to participate.

Pharmaceutical Care Program Study Group

Nine pharmacists were in charge of the pharmaceutical care program. They were trained and certificated in a three-part course. The first reviewed the basis of diagnosis, management and goals of therapy (4 h). The second focused on essentials of communication and simulation sessions of interview and orientation of patients (15 h). In the third step (16 h) the pharmacists were trained in the Dáder method,¹⁵ adapted for use in an outpatient clinic. The Dáder method, which was developed to provide pharma-

ceutical care, consists of seven steps covering the evaluation of services, benefits of interventions, and identification of unexpected occurrences. In addition, in weekly meetings during the investigation, the pharmacists discussed problems identified in the interviews with the consultant physician.

Printed Educational Materials

Printed educational materials about hypertension and treatment with hydrochlorothiazide were developed. In accordance with standardized recommendations,^{16,17} they were modified and validated by physicians, nurses, and patients.

Identification of Drug-Related Problems

The definition of drug-related problem proposed by the Brazilian Consensus of Pharmaceutical Care was adopted¹⁸ and was defined as "any health problem related or suspected to be related to drug therapy that interferes or may interfere in the results of therapy or in the quality of life of the patient." A drug-related problem was diagnosed when rational indication, effectiveness, or safety requirements were not satisfied. The most likely reasons and possible solutions for the occurrence of drug-related problems were discussed with the patient by the pharmacist. In face of more complex situations the pharmacist discussed the problem with the consultant physician. This strategy is part of the Dáder method; because some problems could have serious consequences, the strategy was applied to the control group as well.

Evaluation of Other Characteristics Related to the Knowledge and Use of Drug and Nondrug Treatment

Information about adherence and previous knowledge about drug and nondrug therapies for hypertension was obtained by the Dáder method by means of a pharmacotherapeutic history. Patients were asked to bring their medicines to the appointment. In the presence of the drugs, nine questions about their use were systematically presented: 1) Do you use this medication? 2) Who prescribed it? 3) What is the purpose of the prescription? 4) What are the beneficial effects of this drug? 5) How long are you using this treatment? 6) How many tablets, pills, etc, do you use? 7) How do you use them? 8) How long will you be using this treatment? 9) Have you had any trouble while using this drug? Adherence to the prescription in use was estimated by knowledge about dosage schedules and occurrence of adverse reactions, comparing this information with medical records, rating adherence as satisfactory or unsatisfactory.

Nondrug treatment was evaluated by a generic question about the recommendations to treat hypertension with lifestyle changes. If the patient did not remember one strategy, specific questions were made to help the patient to recall some prescriptions, such as by asking, "And about your diet: what are you doing?"

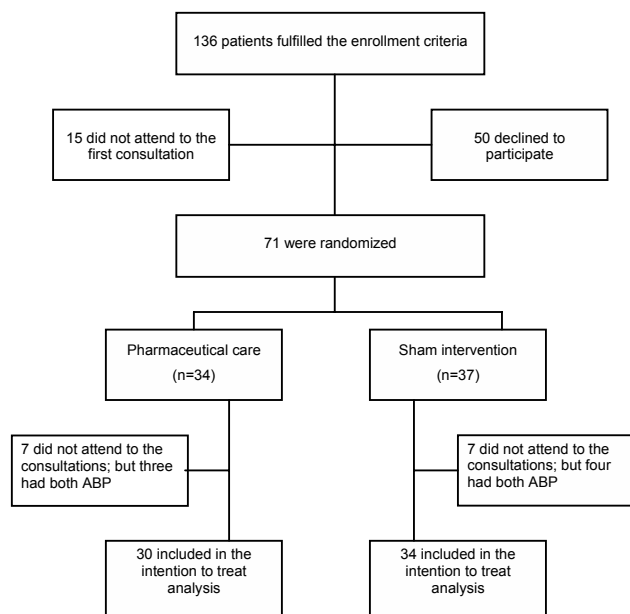


FIG. 1. Flowchart showing allocation of patients to pharmaceutical care and sham intervention. ABP = ambulatory blood pressure.

Table 1. Sequence of procedures of the pharmaceutical care and sham interventions along the trial

Pharmaceutical care program	Sham intervention
First meeting	
Pharmacotherapeutic history	The same
Questionnaire about hypertension	The same
Orientation about hydrochlorothiazide and other drugs (verbal and written)	None
Scheduling of a new meeting in 15 days	The same
Second meeting	
First battery of cognitive tests	The same
Evaluation of the learning process; presentation of the printed material	None
Identification and solution of drug-related problems	None
Scheduling of a new meeting in 1 month	The same
Third meeting	
Second battery of cognitive tests	The same
Evaluation of intervention in pharmaceutical care process	None
Scheduling of a new meeting in 2 month	The same
Fourth meeting	
Third battery of cognitive tests and anxiety	The same
Requesting of ABP monitoring	The same
Scheduling of a new meeting in 2 month	The same
5th meeting	
Fourth battery of cognitive tests	The same
Presentation of ABP monitoring and cognitive results	The same
Blood sample for hydrochlorothiazide determination	The same

Control Intervention

Patients allocated to the control group were submitted to a sham intervention aiming to blind the real objective of the study. It consisted of a series of cognitive tests: Mini-Mental,¹⁹ Self-Reporting Questionnaire-20,²⁰ and Future Self-Perception Questionnaire.²¹ Drug related-problems were also investigated in this group. The pharmacists were trained to apply the questionnaire (10-h course). Table 1 summarizes the activities of patients at each visit in the intervention and control group.

Outcomes

The primary outcome was change of BP measured by ABP monitoring (SpaceLabs 90702 device (SpaceLabs Medical

Inc., Redmond, WA). The examinations were performed just after randomization and in a 7-day period after the intervention. A large cuff was used on patients with an arm circumference >32 cm. The occurrence of drug-related problems was evaluated as a secondary outcome.

Adherence was determined by measurement of plasma levels of hydrochlorothiazide, which was prescribed for all participants. Patients were not informed that adherence would be checked by this measurement. This medication is used as first-line therapy in almost all patients in our clinic and is freely available in the Brazilian public health system. Blood sampling was collected at the last consultation to determine plasma levels of hydrochlorothiazide. The methodology for hydrochlorothiazide quantification was validated according to Register no. 899 of National Agency of Sanitary Vigilance and the ICH-Q2A and ICH-Q2B guidelines.²² Certified pharmacists performed all tests. The test was not able to discriminate between levels of hydrochlorothiazide <20 ng, and therefore we identified only the presence of drug in plasma as an indicator of adherence.

Measurement of Other Baseline Covariables

The protocol used in the outpatient Hypertension Clinic included the assessment of anthropometric variables, number of years of school, presence of co-morbidities, past use of BP-lowering drugs, and other characteristics that were used to compare the experimental groups.

Statistical Analysis

Sampling size was calculated on the basis of a standard deviation of 8 mm Hg in 24-h diastolic BP, observed in a previous trial in our clinic²³; an effect size of 5 mm Hg in 24 h diastolic ABP (the primary outcome); and an α -error of 5% (two-tailed). A sample size of 33 patients per group was estimated to provide 80% of statistical power to reject the null hypothesis.

Within each group, changes (δ) in 24-h, daily, and nightly ABP were calculated by subtracting the baseline values from the values measured after the intervention period. Between-group differences were calculated by subtracting the change observed in the intervention group from the change observed in the control group. Analyses were conducted on intention-to-treat basis. Between-group differences were determined by Student *t* tests for independent samples and the corresponding 95% confidence intervals were calculated. Analysis of covariance was used to adjust the data for baseline BP and age.

Results

About one half of patients screened fulfilled the enrollment criteria and agreed to participate in the study. Of the 71 patients randomized, 57 (80.3%) attended all consultations. Among the 14 participants who did not attend to the

Table 2. Baseline characteristics of participants allocated to the pharmaceutical or sham intervention

Characteristic	Pharmaceutical care (n = 30)	Sham intervention (n = 34)	P
Female	21 (70)	21 (62)	.66
Age (y)	63.9 ± 9.0	59.1 ± 10.1	.052
Body mass index (kg/m ²)	28.8 ± 4.0	30.5 ± 5.1	.17
Duration of hypertension ≥10 years	25 (83)	25 (74)	.261
Uncontrolled hypertension >2 years	19 (63)	22 (65)	.558
Years at school			
0-4	12 (40)	12 (35.5)	.345
5-8	12 (40)	10 (29)	
≥9 y	6 (20)	12 (35.5)	
Comorbidities	3.9 ± 1.9	3.5 ± 1.6	.401
Office systolic blood pressure	169 ± 19	162 ± 15	.138
Office diastolic blood pressure	90 ± 12	90 ± 12	.883
Diabetes mellitus	12 (40)	12 (35.5)	.897
No. of drugs in use	6.7 ± 2.5	5.9 ± 2.2	.187
No. antihypertensive drugs in use	3.0 ± 1.0	2.7 ± 0.9	.192
Previous knowledge about medications			
Satisfactory	14 (46.7)	21 (61.7)	.337
Unsatisfactory	16 (53.3)	13 (38.3)	
Previous knowledge about nondrug therapies for hypertension			
Satisfactory	25 (83.3)	26 (76.5)	.440
Unsatisfactory	2 (6.7)	3 (8.8)	
Did not inform	3 (10.0)	5 (14.7)	
Concordance between information from patient and medical records	20 (66.7)	22 (64.7)	.921

Data are mean ± SD or n (%), when appropriate.

consultations, seven underwent the initial and final ABP monitoring and were included in the intention-to-treat analysis (Figure 1).

Table 2 presents the baseline characteristics of the treatment groups. Participants allocated to pharmaceutical care were older than participants in the control group. The remaining characteristics were similar among groups.

After 24 weeks of follow-up, BP determined by ABP monitoring decreased in both groups in all daily periods

(Table 3). After adjustment for baseline BP and age, decrements in systolic and diastolic BP were slightly higher in the pharmaceutical care group but did not reach statistical significance.

A large proportion of participants experienced a white-coat phenomenon, both in the pharmaceutical care group (46.7%) and in the control group (38.2%). Analysis excluding these participants did not substantially change the results.

Table 3. Ambulatory blood pressure before and after intervention, with respective crude and adjusted Δ -values between change in blood pressure in intervention and control groups

Blood pressure	Group	Before	After	Delta	Δ adjusted (IC 95%)	P*
24-h Systolic	Intervention	140 ± 18	134 ± 11	5	3 (−1 to 5)	.308
	Control	136 ± 14	135 ± 15			
Daily systolic	Intervention	143 ± 19	138 ± 12	4	2 (−2 to 4)	.485
	Control	139 ± 14	138 ± 15			
Nightly systolic	Intervention	135 ± 20	127 ± 13	7	5 (−1 to 6)	.132
	Control	129 ± 16	128 ± 17			
24-h Diastolic	Intervention	80 ± 11	77 ± 10	1.7	1 (−1 to 3)	.541
	Control	79 ± 10	78 ± 11			
Daily diastolic	Intervention	82 ± 12	81 ± 10	1	0 (−2 to 2)	.953
	Control	82 ± 10	81 ± 11			
Nightly diastolic	Intervention	74 ± 11	70 ± 11	4	3 (−1 to 4)	.162
	Control	71 ± 10	71 ± 12			

Data are mean ± DP.

* Analysis of covariance, adjusted for correspondent baseline blood pressure and age.

Table 4. Drug-related problems associated with blood pressure-lowering drugs

Type of problem	Pharmaceutical care program (n = 21)	Sham intervention (n = 20)	P
Indication	7 (11.0)	4 (6.0)	.22
Effectiveness	17 (25.5)	8 (11.5)	.01
Safety	13 (19.5)	17 (25.5)	.60
Total	37 (56.0)	29 (44.0)	.12

Data are n (%).

Office systolic BP decreased by 17 ± 20 mm Hg in the pharmaceutical care group and 12 ± 19 mm Hg in the control group. The corresponding values for diastolic BP were 10 ± 10 mm Hg in the intervention group v 6 ± 14 mm Hg in the control group ($P = \text{NS}$ in both cases).

Hydrochlorothiazide was detected in the plasma of 21 of 27 patients in the intervention group who attended all appointments and in 24 of 30 patients in the control group ($P = .904$). The proportion of patients with satisfactory adherence by history did not differ between the groups. Almost the same number of patients had their prescription changed during the trial: 32% in the control group and 40% in the intervention group ($P = .50$).

Drug-related problems were identified in 50 patients (78.1%); of these, 41 (64.1%) were associated with BP-lowering drugs (Table 4). Seven occurrences were characterized as adverse events. A total of 31 drug-related problems of the 37 detected in the intervention group received a specific pharmaceutical intervention. Lack of effectiveness was more frequently identified in the pharmaceutical care group. In all, 29 drug-related problems were observed in the control group. Seven required intervention, as they were associated with potential morbidity.

Discussion

In this randomized controlled clinical trial, 24 weeks of the pharmaceutical care program led to reduction in ambulatory BP in patients with uncontrolled hypertension that was not statistically different from the reduction observed in the group who underwent a sham intervention.

The unexpected high frequency of the white-coat phenomenon among the participants who fulfilled the enrollment criteria may have reduced the benefit of the active intervention, as several participants had ABP within normal range and could not have their BP decrease further. In planning the trial we tried to reproduce the clinical scenario when we identify patients with uncontrolled clinic BP, that is, we aimed to implement some measures to improve adherence to treatment and to achieve better BP control. Use of ABP monitoring may be helpful in reducing a white-coat effect in this context, but this is not the case in most clinics worldwide and in our country in particular. We therefore used ABP monitoring in our study

to measure more precisely the effect of intervention (performing both a pre-trial and a post-trial examination) rather than using it as a diagnostic tool. The high proportion of patients with the white-coat phenomenon was unexpected, and it limited the power of our study. Another aspect of study design that may have minimized the efficacy of the pharmaceutical intervention relative to the control was the greater-than-usual intensity of the control intervention. Independent of randomization to intervention or control, all participants had the same number of visits to the clinic, and the consultation lasted almost the same time in both groups. Patients from the control group were also frequently asked about the occurrence of drug-related problems to identify any potentially serious adverse event. Despite these limitations, we identified a trend of a beneficial effect of the intervention, particularly during the night-time period. The adjusted difference of 5.1 mm Hg in the systolic ABP between the intervention and control groups during the night would be clinically relevant; however the study was not powered to detect it.

Pharmaceutical care improves the effectiveness of treatments mainly by correcting or preventing drug-related problems. Not all patients in this trial had drug-related problems, and therefore not all would benefit from the intervention. Otherwise, to be beneficial, the practice of pharmaceutical care needs to be offered to all patients with low response to medical therapies, and it is impossible to know a priori which patients would have drug-related problems. The orientation used to correct drug-related problems in the pharmaceutical intervention in this trial may have not been the most efficacious approach. Tailored programs of care and simplification of treatments have been recommended.^{24,25}

The proportion of patients with adherence to hydrochlorothiazide treatment was high in both groups, but the absence of determination of plasma levels of hydrochlorothiazide before the intervention precluded the evaluation of changes in adherence in both the control and intervention groups. The long duration of hypertension, the a priori motivation of patients to participate in the trial, and the free supply of the drug by the Brazilian Health Care System may be some reasons for this high rate of adherence. This rate may not be applied to other medications, as the average number of BP-lowering drugs that usually in use was almost three, and some of these are not freely available in the health care system.

In conclusion, we demonstrate that a pharmaceutical care program, although it did not significantly modify ABP, was feasible and showed a trend for better BP control in patients with uncontrolled hypertension. Further studies are needed to confirm the effectiveness of this approach and to evaluate the performance of different strategies of pharmaceutical care.

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